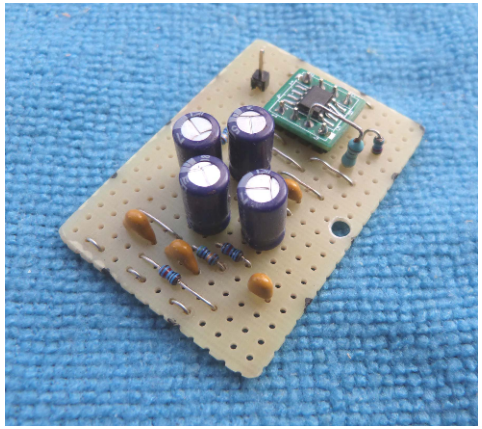
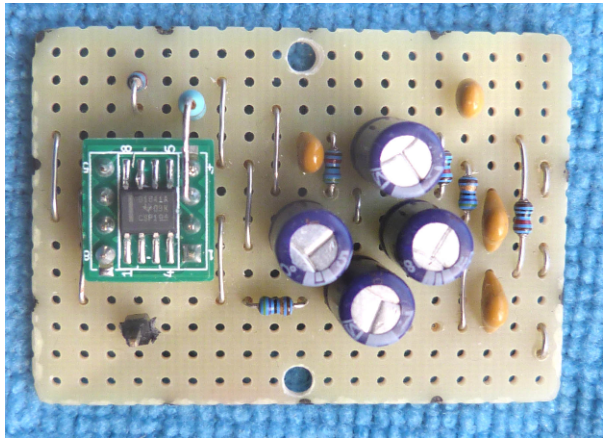
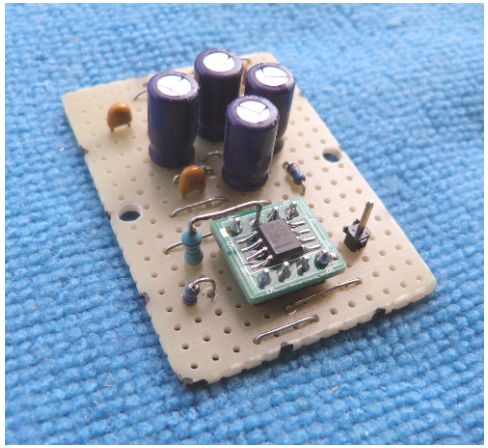
OPIC.46 – Impedance Converter for 'FET-less' Electret Capsule

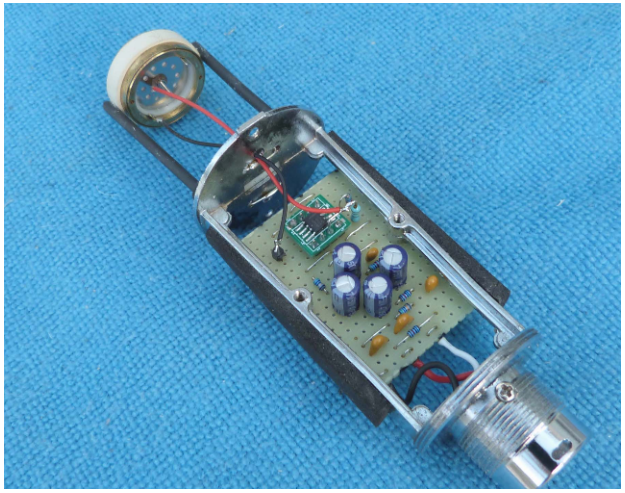
- Simplified Stripboard Layout
- Designed to fit into a BM800 donor body

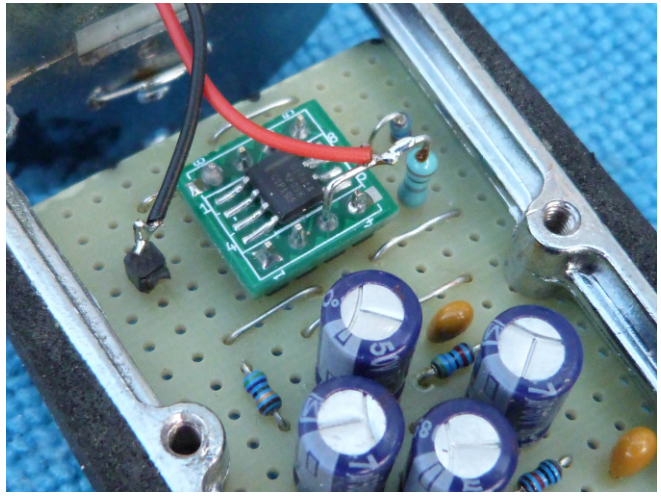
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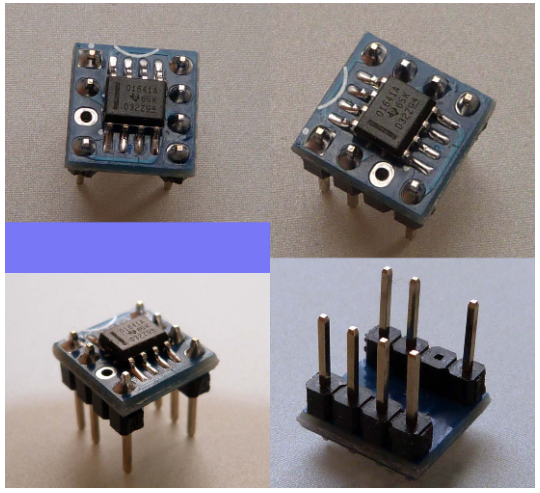




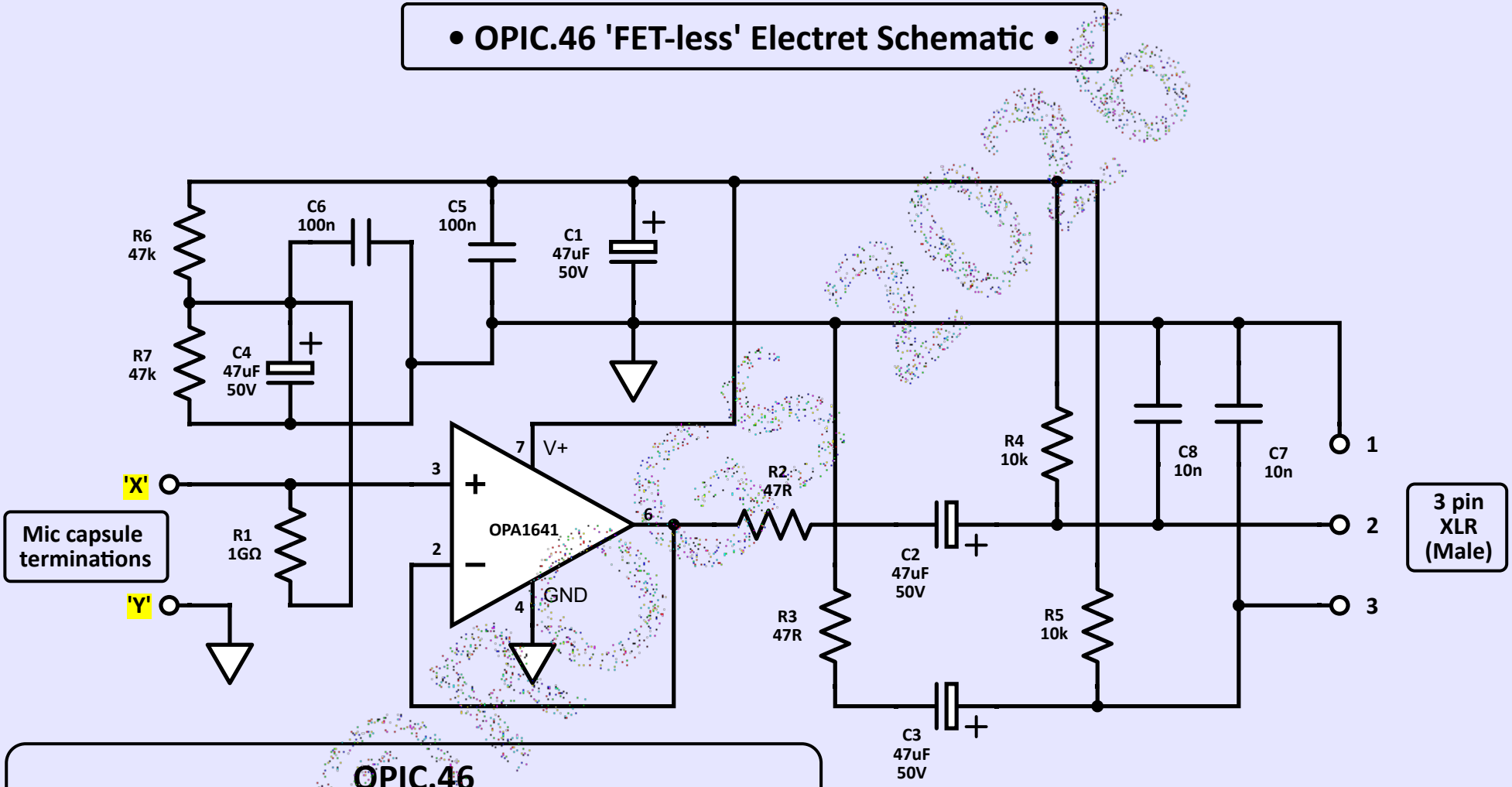








• OPIC.46 'FET-less' Electret Schematic •



OPIC.46

This version is for an electret capsule without an internal FET

- The 'X' connection is made to the capsule centre pin (the + ve terminal)
- The 'Y' connection is made to the capsule case (the - ve terminal)

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OPIC.46 - Circuit description

Overview:

The circuitry shown in the schematic uses a Texas Instruments OPA1641 JFET op-amp as an impedance converter for a 'FET-Less' electret condenser microphone capsule. (i.e. an electret capsule *without* an internal FET).

This op-amp configuration offers a simple alternative to the more conventional use of a discrete JFET device to achieve this function.

The TI OPA164* series of op-amps offer low noise, very low distortion and low quiescent current, making them ideal for this task.

A single OPA1641 op amp is used here to provide a single sided audio output, which is passively balanced to help optimise the common mode rejection ratio (CMRR).

Circuit description:

The OPA1641 is configured as a zero gain non-inverting buffer. The op-amp output is connected directly to the inverting input to achieve this.

The amplifier draws around 1.8mA of quiescent current, and this DC is provided by the 48v phantom power supply from the mic preamp.
The resistor summing network R4 and R5, together with the decoupling capacitors C1 and C5, will allow a smoothed supply voltage of around 24v to be presented to pin 7 of the op-amp.
The actual supply voltage is not critical, as the circuitry will automatically adjust the system functions to suit the supply voltage.

The op-amp requires a 'dual' voltage supply, and this is provided by the creation of a 'half rail' voltage of around 12v by the voltage divider R6 and R7, together with the decoupling capacitors C4 and C6.
The 'half rail' voltage is required to bias the op-amp, so that the output can swing symmetrically around this reference voltage.

The half rail reference DC is fed to the non-inverting input of the op-amp via R1, a very high value 1GΩ resistor.

FET-Less electret capsules only typically have a capacitive value in the order of around 40 or 50pF, and the capsule is connected directly to the non-inverting input of the op-amp. The capsule, in conjunction with R1, forms a high pass filter at the non-inverting input. The very high value of R1 (1GΩ) is chosen so that a full audio frequency range is available at the input, even with the very small capacitive value of the capsule.

The FET-Less capsule used in this version does not require an externally generated polarising DC voltage. The electret capsule is ‘pre-polarised’ as part of the manufacturing process.

The 'X' input connection on the schematic is connected to the capsule centre pin terminal (the +ve terminal), which is connected in turn to the junction of R1, and pin 3 of the op-amp.
This leg of R1 is connected to the vacated pin3 of the op-amp adaptor (see the construction photos on page 1) and is *not* connected directly to the strip-board. The connection between R1/pin3 and the capsule centre pin is deliberately made 'in the air' above IC1, to keep this *very* high impedance point away from flux or other contamination, which can cause noise over time.

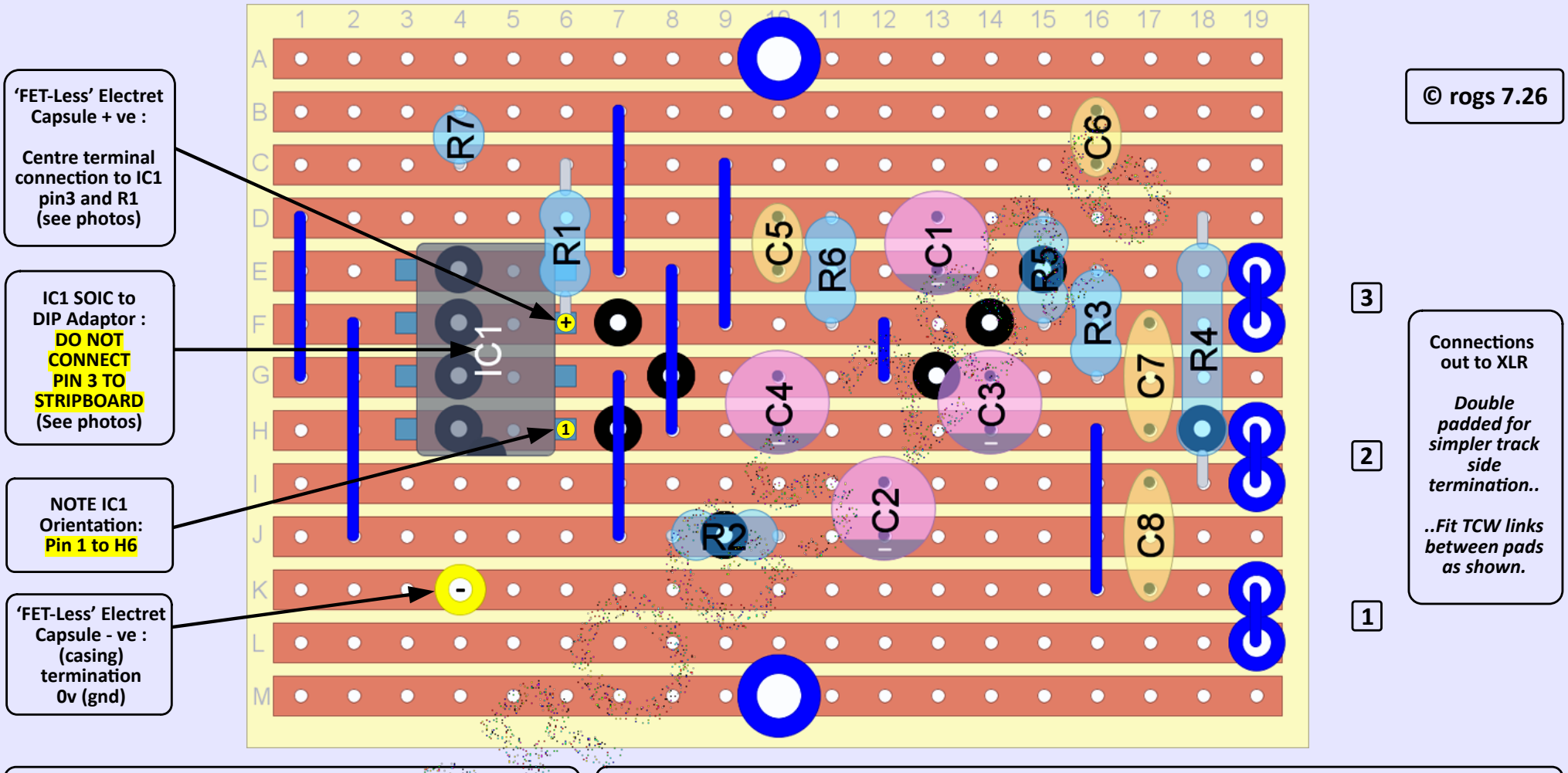
The ‘Y’ capsule terminal (the -ve terminal) is connected to 0v (ground).

The op-amp output (pin 6) is connected via R2 and C2 to pin 2 (hot) of the XLR connector. C2 is required to isolate the DC half rail voltage present on the output of the op-amp from the XLR output.

R3 and C3 provide an equivalent 'passively balanced' output to pin 3 (cold) of the XLR connector. There is no audio on this pin.
R3 and C3 are fitted to improve CMRR, which can be important when the mic is connected to the preamp via a long cable run.

C7 and C8 are included to decouple any stray RF interference which may be present.

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OPIC.46 – Stripboard

• VIEWED FROM COMPONENT SIDE •

- FC2 or FC4 (KEMO E005 or similar)
- 13 (copper tracked) rows
- 19 columns
- A10 and M10 are PCB mounting holes
- 11 x TCW25 wire links (blue lines)
- 12 x 'Spin off' track cuts (black and white dots) (Do not forget the track cuts under R4 and R5)

Fit the components listed below:

Capacitors:
- C1 47uF 50V elect.
- C2 47uF 50V elect.
- C3 47uF 50V elect.
- C4 47uF 50V elect.
- C5 100n 0.1" mlcc
- C6 100n 0.1" mlcc
- C7 10n 0.2" mlcc
- C8 10n 0.2" mlcc

(C5-C8 are COG 63V mlcc)

Resistors:
All resistors (except R1) are MF12 (1/8th watt)

- R1 1GΩ (10%) (RGP0207CHK1G0 or similar)
- R2 47R
- R3 47R
- R4 10k
- R5 10k
- R6 47k
- R7 47k

Semiconductors:
- IC1 OPA1641 (via adaptor) (Pin1 - H6)